

Classifying Gas Distribution Pipeline Incident Causes Using RNN, GRU, and BiLSTM

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Project Purpose

Gas distribution incidents can result from excavation damage, equipment failure, corrosion, natural forces, incorrect operation, and other causes. Each incident may include a written narrative explaining what happened. Manually reviewing and categorizing many such reports can be time-consuming.

We propose to build an NLP-based text classification system that reads a gas pipeline incident narrative and predicts its underlying cause. The project will compare recurrent neural-network models to determine how well they classify technical incident descriptions into the correct cause category.

Data Source

We will use the publicly available **PHMSA Gas Distribution Incident Data, January 2010–Present**, published by the U.S. Department of Transportation Pipeline and Hazardous Materials Safety Administration (PHMSA).

Dataset: [PHMSA Pipeline Incident Data](#)

The downloaded dataset contains **1,592 incident records** and **468 fields**. Of these, **1,582 records contain a written incident narrative**, while all **1,592 records contain an official cause label**. For this NLP task, the main fields will be **NARRATIVE**, which provides the text input, and **CAUSE**, which provides the correct label.

The dataset contains eight cause categories: Excavation Damage, Other Outside Force Damage, Incorrect Operation, Natural Force Damage, Pipe/Weld/Joint Failure, Equipment Failure, Corrosion Failure, and Other Incident Cause. These official categories will serve as the **gold labels**.

Proposed Approaches and Team Contributions

All team members will contribute equally to data preparation, text cleaning, train/validation/test splitting, and final analysis.

- **Parva:** Train an **RNN** text classifier as the baseline model.
- **Muhammad:** Train a **GRU** classifier and compare it with the RNN.
- **Sakina:** Train a **BiLSTM with attention** classifier.

All models will use the same data splits for a fair comparison. We plan to use pretrained **GloVe embeddings** to represent words numerically. Because the cause categories are imbalanced, **class-weighted loss** may be used to reduce bias toward the larger classes.

Evaluation and Final Deliverable

We will evaluate the models using **precision, recall, and F1-score** for each cause category, together with **macro- and micro-averaged** scores and accuracy.

The final deliverable will be a trained NLP system deployed as a **Streamlit web application**. A user will enter a new gas pipeline incident narrative, and the application will use the best-performing model to predict its most likely cause category.

Computational Resources and Deployment

Development will use **VS Code, Git/GitHub, Python, PyTorch, and scikit-learn**. **Google Colab** may be used if additional GPU resources are required. The final Streamlit application will be deployed on **Oracle Cloud Infrastructure (OCI)** for live web access.